Ignition Perspective Screen Design Standard

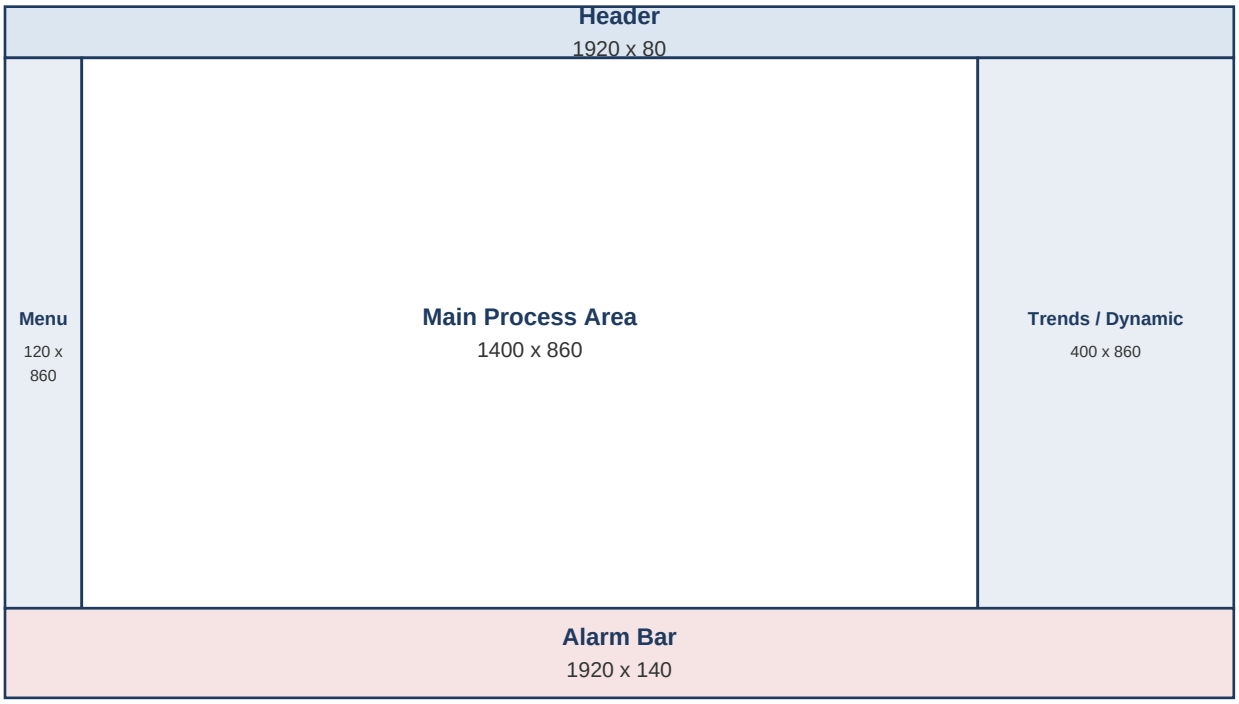
16:9 Scalable Layout & Text Scaling — Final Decisions

Platform: Ignition Perspective 8.1 | Target displays: 1920x1080 through 3840x2160

1. Approach

All operator screens are built inside a single **Shell view** designed at 1920x1080. The shell's root is a Coordinate container in **Percent** mode with the aspect ratio locked to **16:9**, so the entire interface scales proportionally and letterboxes on non-16:9 displays. Each screen region (header, menu, main area, trends, alarm bar) is an Embedded View placed in the shell. Perspective docks are not used for these regions, because docks are sized in fixed pixels and would not stay aligned with the scaled content.

2. Shell Layout



Region	Design size (px)	x	y	width	height
Header	1920 x 80	0	0	100%	7.41%
Menu	120 x 860	0	7.41%	6.25%	79.63%
Main process area	1400 x 860	6.25%	7.41%	72.92%	79.63%
Trends / dynamic window	400 x 860	79.17%	7.41%	20.83%	79.63%
Alarm bar	1920 x 140	0	87.04%	100%	12.96%

Percent values are the position properties to enter on each Embedded View in the shell's percent-mode root. The trends / dynamic window is 400 design px wide, which renders at 800 px on a 3840x2160 display (its largest size) and scales down proportionally on smaller displays. If a collapsible-trends variant of the shell is added later, its main area becomes 1800x860 and should use layouts

designed for that size.

3. Container Settings

View	defaultSize	Root container	aspectRatio
Shell	1920 x 1080	Coordinate, mode = percent	16:9
Region views (header, menu, trends, alarm bar)	Equal to the region's slot size	Coordinate, mode = percent	Blank
Process screens (main area)	1400 x 860	Coordinate, mode = percent	Blank
Templates	The size they occupy on the screen	Coordinate, mode = percent	Blank

Only the shell locks the aspect ratio. Because the shell fixes the overall proportions, every slot keeps a constant shape; setting an aspect ratio on inner views would cause double letterboxing and gaps.

4. Navigation and Dynamic Content

Main area navigation. One page route, `/screen/:name`, points to the Shell as its primary view. The Shell has a `name` view parameter, and the main-area Embedded View's path uses the expression binding below. Menu buttons navigate to routes such as `/screen/Boiler`, so browser back, bookmarks and direct links all work while the header, menu and alarm bar remain in place.

```
"Screens/" + {view.params.name}
```

Trends / dynamic window. The right-hand Embedded View's path and params are bound to session custom properties (for example `session.custom.trend.view` and `session.custom.trend.params`). Clicking equipment on a process screen writes to these properties to change the panel's content. A message handler on the shell (`system.perspective.sendMessage`) is an acceptable alternative.

Menu. At 120 design px wide (about 80px on a 1280x720 display), the menu uses icons with a short one-word label beneath each.

Alarm bar. At 140 design px tall, the alarm bar shows roughly four alarm rows plus a header row at about 28 design px per row.

5. Text Scaling Standard

All text, padding, borders, border radius and icon sizes are sized in **design pixels** (dp): one pixel on a 1920x1080 display, scaled to the current screen. The unit is defined once as a CSS variable in a custom theme's variables file, or in the Advanced Stylesheet if the installed 8.1 build includes it:

```
:root {
  --dp: min(0.052083vw, 0.092593vh);
}
```

Any size is then written as `calc(N * var(--dp))`, where N is the size in pixels on a 1080p display. Because the unit is derived from the viewport rather than from the enclosing container, it gives the same result at any nesting depth: text in a template inside an embedded view inside the shell scales at exactly the same rate as

the shell itself. Where the variable is unavailable, the equivalent direct form is `min(N/19.2 vw, N/10.8 vh)`.

Text sizes are applied through Style Classes rather than set on individual components:

Style class (suggested)	1080p size	fontSize using --dp	Direct equivalent
Text/Small	12	calc(12 * var(--dp))	min(0.625vw, 1.111vh)
Text/Label	14	calc(14 * var(--dp))	min(0.729vw, 1.296vh)
Text/Body	16	calc(16 * var(--dp))	min(0.833vw, 1.481vh)
Text/Header	20	calc(20 * var(--dp))	min(1.042vw, 1.852vh)
Text/Title	28	calc(28 * var(--dp))	min(1.458vw, 2.593vh)

Line height is set as a unitless value (for example 1.2) so it follows the font size. Fixed `px` values are not permitted in any font size, padding or border width on operator screens.

6. Resolution Behavior

Element	1920 x 1080	3840 x 2160
Shell	1920 x 1080	3840 x 2160
Main process area	1400 x 860	2800 x 1720
Trends / dynamic window	400 x 860	800 x 1720
A 200 x 120 template	200 x 120	400 x 240
Text at calc(14 * var(--dp))	14 px	28 px
Text at a fixed 14px (not permitted)	14 px	14 px (half the relative size)

4K monitors running Windows display scaling (150% or 200%) report a smaller CSS viewport to the browser. The dp unit handles this automatically; no per-scaling configuration is required.

7. Templates

Each template is designed at the size it will occupy on the screen, with a percent-mode Coordinate root and no aspect ratio. All template text uses the dp style classes. Templates are placed at their designed size so text stays proportional to the graphic, which also supports consistent symbol sizing across screens.

When the same template must appear at a different size, it takes a `scale` parameter (default 1) and its font sizes are bound with an expression such as:

```
"calc(" + (14 * {view.params.scale}) + " * var(--dp))"
```

8. Components With Internal Pixel Sizing

Several components set fonts, row heights or padding internally in pixels and must be overridden explicitly with dp values: the Table and Alarm Status Table (header style, row style and row height), the Power Chart and Time Series Chart (axis and legend fonts), and the Menu Tree, Dropdown and Tab Container (font and padding). Icons are sized in dp so they stay proportional to adjacent labels. Images use `fit.mode = contain`, and SVG graphics are preferred over raster images.

9. Known Limitations and Testing

The Designer evaluates vw and vh against the whole workspace panel, so text sizing appears incorrect there and must be judged in a browser session. Readability at the smallest display is the limiting factor; the smallest routinely used design size should be 14 dp or larger, and any floor applied with `max(10px, ...)` must be checked for text overflow.

Before rollout, a test screen containing a template at two sizes, a table and a chart is opened in Chrome with the device toolbar set to 1920x1080 and 3840x2160 (plus a non-16:9 size such as 1920x1200 to confirm letterboxing). Text should occupy the same proportion of its box at every size; any element that does not still has a fixed pixel value or an unset component-internal size.

These techniques carry forward unchanged to the planned Ignition 8.3 upgrade.